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Application of XGBoost and artificial neural networks in predicting housing project productivity

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Abstract

Despite its challenges, the building sector remains a crucial pillar of Iraq's economic progress. This study aims to develop and evaluate machine learning models for predicting the productivity of housing projects in Iraq based on project design and implementation characteristics. Three advanced prediction techniques were used, including artificial neural networks (ANN), extreme gradient boosting (XGBoost), and support vector regression (SVR). Actual data from 67 housing projects were used, including seven input variables and one output variable represented by implementation productivity (real time). The evaluation results showed that the XGBoost model achieved the highest predictive accuracy with a determination coefficient of $R^2 = 0.998$ and a relative error (MAPE) of 0.8%. In comparison, the ANN model also performed well ($R^2 = 0.92$ and MAPE = 4.1%). In contrast, the SVR model performed less well with an accuracy of $R^2 = 0.813$ and MAPE = 6.6%. The results of the feature importance analysis also revealed that the most influential factors in productivity were built-up area, followed by the quantities of brick and concrete works. It can be implied that machine learning techniques, particularly XGBoost and ANN, provide effective and accurate tools to support decision-making in the early stages of housing project planning by predicting executive performance based on project data.

Keywords Construction productivity, Artificial neural network (ANN), Extreme gradient boosting (XGBoost), Support vector regression (SVR), Machine learning, Housing projects

1 Introduction

The construction sector is one of the main pillars of economic development in countries around the globe. It plays a significant role in boosting gross domestic product (GDP) in both developed and developing countries. Despite the sector's increasing reliance on technological innovations, it continues to depend heavily on human labour, which results in performance disparities and heightens the likelihood of delays, cost overruns, and poor execution quality [1]. Numerous studies indicate that construction projects face recurring challenges, including administrative corruption, schedule delays,



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budget overruns, and poor executive performance [2]. Despite significant investments and growing interest in construction projects, the sector continues to face fundamental problems related to low labour productivity and high costs. Hence, work productivity efficiency is a decisive factor in effectively managing construction projects [3].

Declining construction productivity (CP) often leads to project delays and unexpected cost increases. This phenomenon has been documented globally, with estimates suggesting that construction productivity losses range from 40 to 60% worldwide. For instance, labour productivity in the US construction sector has decreased steadily for decades, in contrast to other industrial sectors [4, 5]. Furthermore, labour costs account for 30% to 50% of the total project cost in many countries, making labour productivity one of the key indicators in determining project profitability [6, 7]. This issue is especially important in the Iraqi context due to rapid population growth and a severe housing shortage. The Organisation for Economic Co-operation and Development (OECD) has stated that the construction sector is directly linked to population growth rates [8].

According to the annual statistical report of the Ministry of Planning, the population of Iraq in 2021 surpassed 41 million, indicating a growth rate of 3.3%. The capital city of Baghdad has the largest population concentration, with almost nine million people, making up 21.95% of Iraq's total population. It is estimated that the population of Iraq will exceed 70 million people by 2050 [9].

According to the report released by the UNESCO Office in Baghdad, the construction sector and building industry are major contributors to the economy of Iraq [10]. Numerous construction projects, including residential complexes, highways, bridges, hospitals, and universities, are now under renovation. The majority of these developments were allocated to residential complexes [11]. In spite of the existing residential complex developments, they poorly fulfil the market demands and accommodate the significant population growth. In 2020, the housing deficit was estimated at three million housing units [12]. To accommodate the high population growth, it is imperative to construct housing units at a corresponding rate. Over time, various factors and constraints that influence the implementation of residential complexes have been identified, with construction productivity being the most significant concern [13]. Nonetheless, the construction sector in Iraq faces specific challenges, including political instability, economic fluctuations, and resource constraints, necessitating the adoption of advanced technologies to enhance productivity [14]. In this context, machine learning technologies are considered a promising solution due to their ability to analyse big data and extract complex relationships between inputs and variables that affect executive performance.

This study aims to improve the productivity of housing projects in Iraq using artificial intelligence techniques. It focuses on the duration of construction work (number of days) as a practical indicator of productivity. It is assumed that a shorter duration, with a constant amount of work, reflects a higher level of productivity. This indicator is measurable using actual field data. The researchers focus on employing and comparing three modern predictive techniques within the machine learning framework: artificial neural networks (ANN), extreme gradient boosting (XGBoost), and support vector regression (SVR), to predict the productivity of housing projects in Iraq. Real data was collected from horizontal housing projects, including design and implementation variables. The study emphasizes structural activities in housing projects, such as foundations, load-bearing walls, and roofs. By building predictive models using machine learning

techniques, the study seeks to develop decision support tools that assist contractors, engineers, and planners in accurately estimating productivity during the early planning stages. These models will enhance resource management and minimize time and financial waste, thereby improving the efficiency of the housing sector in Iraq and contributing to alleviating the growing housing deficit.

2 Literature review

2.1 Construction productivity

Construction productivity is a key issue in the construction sector, as it reflects the relationship between inputs such as materials, labour, and equipment, and the outputs resulting from these inputs [15]. Many studies have been conducted to define productivity, develop methods to measure and improve it, and analyse the factors that influence it.

In Malaysia, Chia et al. [16] studied the relationship between construction productivity and economic growth between 1970 and 2011. The results of the partial correlation analysis showed a positive relationship between construction productivity and GDP per capita in the 1985–1998 and 1998–2009 cycles, while this relationship was not found in the 1970–1985 cycle. The study pointed out that economic fluctuations, foreign labour inflows and government fiscal policies are among the most important factors affecting productivity and recommended management improvement of the construction cycle to enhance efficiency during periods of economic expansion and contraction. In Cambodia, Durdyev and Ismail [17] conducted a field survey which contained 30 questions about off-site manufacturing and the extent of the impact of related factors. Seventy-five respondents from the construction sector completed the questionnaire. After analysis, the findings showed that the most influential factors were: planning and implementation methods, project cost, industry culture, skilled labour shortage, supply chain, and tasks at the project site. As for design requirements, Fox et al. [18] examined design compatibility with implementation conditions. Differences in quality and productivity between customer-driven designs and those developed by designers were found as determining factors. In Australia, Rose and Manley showed that special incentive systems contributed to higher performance and productivity and increased employee loyalty to the project [19].

On the other hand, Nguyen et al. [20] examined the effect of working hours (day and night) on the productivity of specific tasks, such as road grading and pile driving. They found that productivity was higher during the day, while other tasks, such as manhole casting, showed higher productivity at night. The study also revealed that working hours were not a determining factor for tasks such as drilling and laying pipe networks. In Egypt, El-Gohary and Aziz [21] analysed a set of factors affecting construction productivity through a questionnaire that included 13 key indicators classified into three groups: labour, industry, and management. The most critical factors were team experience, incentive programmes, availability of materials, management, and supervision. Dabirian et al. [22] highlighted the impact of the number of work teams on site, showing that overcrowding at the work site can lead to a drop in productivity up to 30% compared to normal rates. In a similar study, Moselhi et al. [23] examined the impact of change orders during project implementation on construction performance. They analysed 33 real-life cases in the United States and Canada and developed a mathematical model to calculate the production losses resulting from these changes.

Despite their diversity, these studies can be classified into four main categories: studies that addressed factors affecting productivity, studies that defined productivity and its importance, studies that measured and predicted productivity, and studies that focused on improving productivity.

2.2 Housing sector and construction challenges

The housing sector is multidimensional, as the concept of 'housing' in English literature can be understood as a noun referring to the physical dwelling unit that is built, consumed, sold, and purchased, or as a verb meaning to provide shelter [24]. Housing projects encompass all stages from planning and design to implementation and consist of residential units of various sizes and specifications that meet the population's needs. These projects range from horizontal units (houses) to vertical projects (apartment buildings).

Housing projects are a fundamental pillar of economic and urban development, as they contribute to infrastructure development and promote urban growth [25]. Given the sector's importance, numerous studies have been conducted to assess its performance and analyse its challenges. In Jordan, Kasabreh and Tarawneh [26] evaluated the performance of housing projects by analysing the impact of contractors, using a questionnaire as a data collection tool. The results showed that the most critical factors for contractor success were quality assessment, workforce safety, supervision, and management. Durdjev and Mbachu [27] indicated that the housing sector contributes 8% of gross domestic product in Cambodia. After a thorough literature review and analysis of interviews, they identified 36 factors influencing productivity performance in housing projects and then classified them into four categories: location-related factors, labour, external factors, and inventory. The five most influential factors were weak management, insufficient cash flow, poor execution, lack of skilled labour, and change orders during execution. In the same context, Hromada et al. [28] investigated the total cost of the life cycle of housing projects and concluded that design and environmental aspects significantly impact cost. Although sustainable buildings are more expensive to construct, they are a better economic option in the long run due to their lower operating costs. In Kuwait, Koushki et al. [29] examined the reasons for cost increases and delays in private housing projects through a study of 450 project owners. The results showed that modification requests, financial problems, and inexperienced owners were the leading causes of delays, while increased costs were linked to poor contractor performance. In Iraq, the housing sector suffers from a significant shortage of housing units, estimated at three million units [12]. The researcher attributed this deficit to economic and demographic factors, poor implementation, inadequate maintenance, and limited private sector investment. On the other hand, Hassoon et al. [30] conducted a field study using a questionnaire and opinion polls to analyse factors affecting the productivity of steel structure installation teams.

These studies show that housing projects, like other construction projects, face challenges related to cost overruns, delays, and quality of implementation. Hence, modern technologies and innovative forecasting methods that contribute to improving implementation efficiency and achieving economic growth are needed. This research, therefore, focuses on housing projects and their productivity as a key pillar for addressing the housing shortage and achieving sustainable development in Iraq.

2.3 Applications of machine learning in construction projects

Given the importance of increasing productivity in the construction sector, recent years have seen a marked increase in research aimed at predicting productivity and improving performance through artificial intelligence technologies, particularly machine learning techniques. Researchers have sought to employ these intelligent tools to support project management, improve outputs, and accurately predict future productivity [31]. Vogl and Abdel-Wahab [32] claim that the construction industry's project-based nature is one reason for the lack of comprehensive data on construction productivity. Thomas [33] used key performance indicators to assess labour performance in projects, highlighting the importance of using multiple indicators given the difficulty of relying on a single success indicator to assess labour productivity. In a case study, Khanh et al. [34] developed a Bayesian Network model to simulate worker productivity in construction. The results showed that task complexity, technological integration, and work accidents significantly affect productivity over time. The model's predictive error rate was 6.6% for formwork, 10% for reinforcement, and 9.3% for concrete pouring. Lu et al. [35] used artificial neural network techniques to develop a model that accurately estimates construction productivity. The model was evaluated using data from a local construction company and proved to be superior to traditional models in terms of accuracy and providing a probability distribution of productivity that enhances decision-making. In the same context, Golnaraghi et al. [36] used supervised learning algorithms to estimate the productivity of mould workers, and the results of the study showed that the Bayesian neural network (BNN) outperformed other models in prediction. Ebrahimi et al. [37] also showed that combining neural networks and genetic algorithms improves project performance regarding cost, duration, and quality.

On the other hand, AbouRizk et al. [38] relied on neural networks to develop a framework that estimates productivity through working hours per unit. Ok and Sinha [39] focused on analysing equipment performance using linear neural networks, and the study demonstrated the ability of these techniques to improve the accuracy of equipment productivity predictions. In a study of power plant projects in Iran, Heravi and Eslamdoost [40] developed a model based on multilayer neural networks to estimate worker productivity. The results showed that training stopping and Bayesian regularisation techniques improved the model's performance. The study also emphasised the importance of sensitivity analysis to understand the influencing factors accurately. In the same vein, Goodarzizad et al. [41] developed a hybrid model combining neural networks and the genetic optimisation algorithm (ANN-GOA) to predict the productivity of concrete casting work, where 19 workers were classified into four categories: administrative, economic, technical, and environmental. The study found that the most influential factors were worker experience, incentives, wages, accidents, and poor supervision. The model demonstrated high predictive accuracy.

In a comparative study conducted by Sarihi et al. [42] two prediction models were developed: neural networks and a fuzzy inference system (FIS). Data from 124 projects were used to train the model, and the results showed high accuracy when using the hybrid system, which was evaluated using R, MSE, and MAE indicators. Finally, Tsz et al. [43] reviewed the benefits and challenges of modular integrated construction (MIC) technology, showing that the most notable benefits of this technology are improved productivity, reduced execution time, and increased construction quality and safety.

Recent studies indicate that the XGBoost (extreme gradient boosting) algorithm is one of the most effective machine learning techniques for analysing construction data and predicting productivity. Coffie and Cudjoe [44] developed a model for estimating the investment cost of precast concrete buildings using XGBoost. The results showed that the model outperformed traditional models such as artificial neural networks (ANN), support vector machines (SVM), and random forests (RF) in terms of accuracy and training speed, with an evident ability to interpret influential variables and support investment decisions in the early stages of a project. Yan et al. [45] showed that XGBoost excels at handling non-linear and complex data and is characterized by model interpretability. This makes it an ideal tool for accurately estimating labour productivity and project costs. Thus, the literature confirms that integrating AI techniques, especially ANN, SVM, and XGBoost, into the construction sector improves productivity forecasting, reduces error rates, and supports effective decision-making at various stages of the project life cycle.

Moreover, challenges include a lack of standardised criteria, high initial costs, and supply chain complexities. Through these studies, artificial intelligence technologies, especially machine learning technologies, provide effective and accurate tools to improve the productivity of construction projects and reduce error rates, supporting the growing trend towards adopting these methods in various planning and implementation stages in the construction sector.

2.4 Gaps in literature and research justification

To clarify the position of this study within the current literature, Table 1 provides a comparative summary of recent studies that employ machine learning models in the fields of construction productivity and concrete performance. This table compares the models used, data size, focus area, main strengths, and limitations of each study.

Although many studies have addressed the concept of construction productivity and proposed different models for its estimation and improvement, clear research gaps remain. First, most previous studies have focused on identifying factors that influence productivity or have employed traditional statistical models, such as regression analysis, to predict productivity levels. While these methods provide a preliminary understanding, they often fail to capture the complex and non-linear relationships between project variables. Second, some recent studies have introduced artificial intelligence and machine learning techniques to enhance prediction accuracy, but most have relied solely on a single technique, particularly artificial neural networks (ANNs), without comparing their performance to other modern algorithms. This limitation restricts researchers and practitioners from verifying the effectiveness of these models or generalizing their results. Third, many productivity prediction models have been developed using general or synthetic data, with insufficient focus on the housing sector, especially in developing countries, including Iraq, where implementation conditions, labour behaviour, and financial and logistical constraints differ. Furthermore, few studies have used real housing project data to construct these models. Finally, there is a notable lack of studies that directly compare various machine learning techniques, such as artificial neural networks (ANN), exponential gradient boosting (XGBoost), and support vector regression (SVR), using the same dataset. Such a comparison is essential to determine the most accurate and suitable model for application in real-world construction environments. Based

Table 1 Recent studies on machine learning applications in construction

Author(s)	Model used	Data size	Focus	Strengths	Limitations
Kadam et al. (2023)	Hybrid RF	300+ lab samples	Strength of HVFA SCC	High accuracy	Lab-scale only
Kumar & Patil (2023)	Multiple ML Models	150 mixes	M-sand & FA strength	Material evaluation	No deployment
Yadav & Goyal (2023)	XGBoost & LightGBM	200 mixes	Chloride permeability	Advanced boosting	Limited domain
This Study	ANN & XGBoost	67 real projects	Productivity	Field-based, interpretable	Small sample size
Chen et al. (2022)	XGBoost	400 concrete samples	Slump prediction	Fast inference	Limited interpretability
Zhou et al. (2021)	ANN	120 structural cases	Shrinkage forecast	Simple design	Overfitting risk
Wang et al. (2022)	LSTM	360 time series points	Time-dependent load	Long-term trends	Needs large data
Ahmed et al. (2020)	SVR	250 lab sets	Concrete compressive strength	Good accuracy	Not field validated
Singh & Jain (2021)	RF & SVM	310 samples	Flexural strength	Robust to noise	High preprocessing
Li et al. (2022)	Gradient Boosting	500 concrete data	Compressive strength	Large dataset	Training time
Zhang et al. (2023)	CNN	1000 images	Crack detection	Visual analysis	Data labeling
Nasir et al. (2021)	MLR & ANN	180 records	Curing effects	Combined effects	Small data
Rahman et al. (2022)	Bayesian NN	220 mixes	High-temp resistance	Probabilistic prediction	Limited scalability
Mehta & Patel (2021)	RF	200 sets	Fly ash effect	Performs well on variation	No interpretability
Alavi et al. (2022)	CatBoost	280 samples	Blast furnace mix	Handles categorical vars	Newer model
Sharma et al. (2020)	SVM	150 sets	Fiber content impact	Suitable for small data	Not explainable
Omar & Al-Dulaimi (2023)	XGBoost	130 projects	Construction delays	Real-case data	Regional scope
Park et al. (2021)	Ensemble ANN	200 records	Bridge fatigue	Combined hidden layers	Model complexity
Ali & Abbas (2022)	ANN	170 housing units	Cost forecasting	Practical application	Requires domain tuning
Nguyen et al. (2020)	MLR	140 sets	Concrete setting time	Simple regression model	Linear assumption

on these gaps, this study aims to make a scientific contribution through the following: collecting real field data from housing projects, comparing multiple machine learning models, identifying key influencing variables, and proposing a practical predictive framework.

3 Research methodology

This study employed a quantitative methodology, grounded in supervised machine learning algorithms, to predict the productivity of housing projects in Iraq. The methodology consisted of four main stages: data collection, data preparation and processing, model building using three distinct techniques, and performance evaluation and comparison.

3.1 Data collection and description

Data was collected from 67 landed housing projects implemented in central Iraq between 2020 and 2023, reflecting recent practices in the construction sector and enhancing the applicability of the model in current circumstances. These projects included actual implementation data related to structural works such as foundations, load-bearing walls, and concrete roofs.

The duration of execution in days was adopted as a dependent variable for prediction in this study, representing an indirect indicator of productivity. It is assumed that the shorter the execution time, the higher the construction productivity under the same work condition. These durations were chosen based on the daily work reports of each project.

Essential data for the study were extracted from daily reports to determine the number of work teams and periods needed to complete the structural work for each unit. Other information from the unit's drawings is used to specify the details of each housing unit. Following the data-gathering phase, the data underwent examination and processing to extract the variables required for the mathematical model. Using statistical approaches and domain experience, key aspects that substantially influence construction productivity were selected. This phase guarantees that the model concentrates on the most relevant variables. The study relied on seven independent input variables, as shown in Table 2.

To verify the statistical balance between the training and test data and ensure the fairness of the evaluation, the Kolmogorov–Smirnov (K–S) test was performed on the execution time values [46]. The results showed that the probability value ($p = 0.802$) is greater than 0.05, indicating that there was no significant difference between the two distributions; therefore, the data split was statistically appropriate. This supports the model's reliability during evaluation. The study included 67 projects, with 7 independent variables, resulting in an average of 9.57 samples per variable. According to the common statistical rule that the minimum sample size in regression analysis should be between 5 and 10 observations per independent variable [47], this size is considered within the acceptable range to ensure reliability and statistical significance.

3.2 Data pre-processing

The data was entered into the Orange Data Mining program, and the initial processing included the following steps: converting numerical values using the Continuous tool which makes them interpretable by the models; normalizing the data so that the values fall within a specific range (between -1 and 1); and identifying the independent

Table 2 Input variables

Variables	Description	Unit
X1	Land total area	Sq.m
X2	This area has been built on the land, including the areas covered by the building, such as rooms and corridors	Sq.m
X3	The number of rooms in the residential unit	No
X4	The result of dividing the built area by the total area	/
X5	Quantity of Concrete Work	Cu.m
X6	Quantity of Brick Work	Cu.m
X7	Number of Brickwork teams	no

variables and the target variable using the Select Columns tool. They were normalised using Eq. 1, the input value w , which shows the normalisation [48, 49].

$$X \text{ adjusted normalize} = 2 \left(\frac{X_{\text{act}} - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}} \right) - 1 \quad (1)$$

3.3 Model building

Hyperparameters were selected for the ANN and SVR models based on previous studies, then adjusted and optimised using a trial-and-error method supported by five-fold cross-validation. In the XGBoost model, a manual grid search was conducted for a set of basic parameters, including learning rate, maximum depth, and $n_{\text{estimators}}$, to achieve optimal performance and minimize the risk of overfitting. Three machine learning models were developed using the same data, according to the following settings:

3.3.1 Artificial neural network (ANN)

The artificial neural network (ANN) model was constructed using a hidden layer containing seven nodes, which were activated using the Tanh activation function as shown in Eq. 2 [50]. The L-BFGS-B optimisation algorithm was adopted as the solution method, with a regularisation coefficient (α) of 0.004 to reduce the risk of overfitting. The maximum number of training iterations was set to 130, and the repeatability property was fixed to obtain stable results. The model used a learning rate of 0.1 to adjust the weights during the training process. These parameters were selected based on previous studies [51] and then were optimised using trial-and-error to fine-tune the model and reduce the risk of overfitting, using fivefold cross-validation to ensure the stability of the results.

$$\text{Tanh} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (2)$$

3.3.2 Regression using support vector regression (SVR)

The regression model was developed using a support vector machine (SVR) with a polynomial kernel of degree $d=3$ [52]. The regularisation coefficient C was set to 1.0, and the regression error (ϵ) was set to 0.1 to adjust the permissible error margin. The bias coefficient c was set to 1.0, while γ was set to Auto. The solution accuracy was set to 0.001, and the maximum number of training iterations was set to 100. These parameters were determined following a study conducted by [53] and were manually adjusted to achieve a balance between bias and variance based on the characteristics of the data set used.

3.3.3 Gradient boosting (XGBoost)

The XGBoost model was built using the scikit-learn library, where the number of trees was set to 100 and the learning rate to 0.1. The depth of each tree was set to 3 levels, and the minimum sample size per leaf was set to 2. The proportion of subsamples used in each training round was 1.0, meaning the entire sample was used each time.

All models were connected to the Test & Score tool to evaluate their performance using the same cross-validation settings. The models were evaluated using the following statistical performance indicators: mean square error (MSE), root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), coefficient of determination (R^2), and coefficient of correlation (R) [54]. To ensure a fair

evaluation, fivefold cross-validation was applied. The Scatter Plot tool also illustrates the relationship between actual and predicted values. To reduce the risk of overfitting, several techniques were applied during model development, including data normalization, an 80/20 train-test split, early stopping, and dropout layers in the ANN model, and regularization parameter tuning in XGBoost. These steps ensured balanced learning and better generalization of the models. The accuracy of the models was further validated by entering new project data that was not used in training, utilizing the predictions tool, and comparing the results. Figure 1 shows the methodology flow chart.

4 Results and discussions

Three predictive models were evaluated to estimate the productivity of housing projects in Iraq: artificial neural networks (ANN), the XGBoost gradient boosting algorithm, and regression using a support vector machine (SVR). A set of statistical performance indicators was adopted to compare the accuracy of the models, including mean square error (MSE), root mean square error (RMSE), mean absolute error (MAE), mean absolute

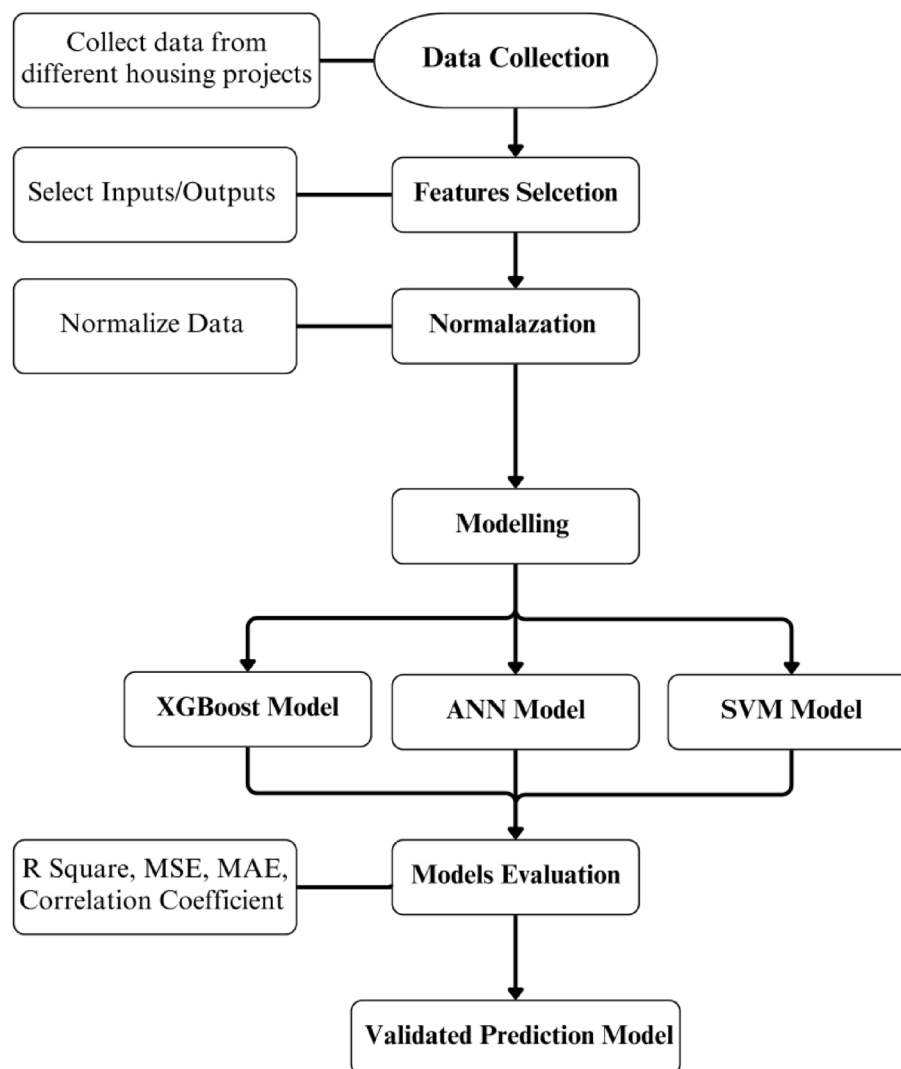


Fig. 1 Research methodology

relative error (MAPE), and coefficient of determination (R^2). Table 3 illustrates the statistical metrics for each model.

The results of the statistical analysis indicate that the XGBoost model was the most efficient in predicting the productivity of housing projects compared to other models. The model achieved the lowest error indicators, with a mean square error ($MSE = 1.33$), root mean square error ($RMSE = 1.156$), and mean absolute error ($MAE = 0.917$). The mean absolute percentage error (MAPE) was also very low (0.80%), in addition to the coefficient of determination ($R^2 = 0.998$) and the correlation coefficient ($R = 0.999$), reflecting very high predictive accuracy within the data used in the study. However, the sample size remains limited, which calls for external verification to enhance the generalisation of the results.

The artificial neural network (ANN) model performed relatively well, achieving a coefficient of determination ($R^2 = 0.92$) and an absolute mean percentage error ($MAPE = 4.1\%$), which are acceptable results demonstrating the model's ability to represent non-linear relationships. However, the error indicators ($MSE = 71.6$, $RMSE = 8.46$) were much higher than those achieved by the XGBoost model, indicating that ANN, despite its effectiveness, requires more precise parameter tuning and may be more sensitive to data distribution.

In contrast, the support vector machine (SVM) model performed the worst, recording the highest error metrics ($MSE = 155.925$, $RMSE = 12.487$, $MAE = 7.863$), in addition to an absolute relative error rate ($MAPE = 6.6\%$) and a relatively weak determination coefficient ($R^2 = 0.813$). This reflects the limitations of this model in representing the complex relationships between housing project variables.

To confirm the statistical differences between the models, a Friedman test was performed using the RMSE values resulting from repeated experiments. The results showed a value of ($\chi^2 = 10.00$) with a probability value of ($p = 0.0067$), indicating a statistically significant difference between the models and confirming the significant superiority of the XGBoost model over both ANN and SVM in predicting project productivity.

These results are consistent with previous studies, as recent research has indicated that the XGBoost algorithm outperforms other machine learning models in predicting productivity in construction projects, achieving determination coefficients exceeding 0.97, which is close to the results of this study. The ANN results also support these trends, with some studies [55] showing that ANN achieves determination coefficients close to 0.95 with acceptable error rates (MAPE less than 5%), which are similar to the results obtained in this study.

Overall, the comparison between the three models shows that XGBoost is the most accurate and efficient model for predicting the productivity of horizontal housing projects in Iraq, followed by the ANN model, which has proven its efficiency in representing

Table 3 Model statistical metrics

Metrics	Models		
	XGBoost	ANN	SVM
MSE	1.33	71.6	155.925
RMSE	1.156	8.46	12.487
MAE	0.917	6.06	7.863
MAPE	0.80%	4.1%	6.60%
R^2	0.998	0.92	0.813
R	0.999	0.96	0.901

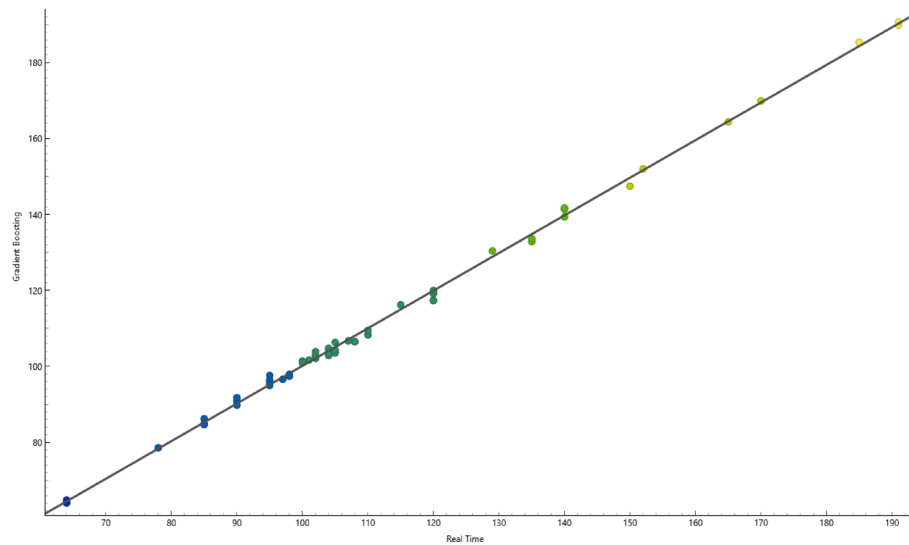


Fig. 2 Scatter plot between actual and expected values using the XGBoost model

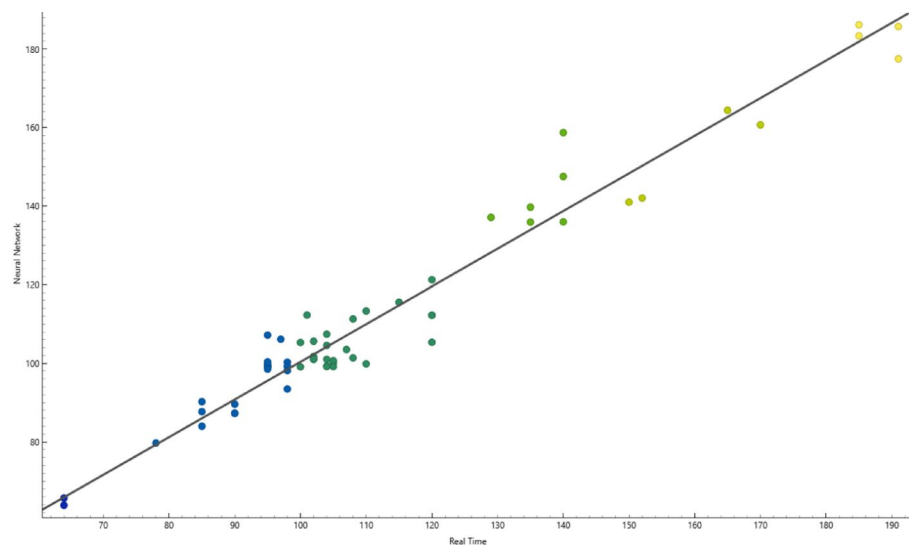


Fig. 3 Scatter plot between actual and predicted values using the ANN model

non-linear relationships, while SVM performed less effectively. These results highlight the importance of using modern machine learning techniques, especially XGBoost and ANN, as promising tools to support decision-makers in the early stages of housing project planning, provided that the database is expanded and additional validation tests are applied to enhance the reliability of the models. Figures 2 show the scatter plot results of the XGBoost model.

Figure 3 shows the result of a scatter plot for the ANN model.

Figure 4 illustrates the scatter plot of SVM model.

The scatter plots reinforce the conclusion that the XGBoost model is the most effective and reliable. The artificial neural network (ANN) model follows with acceptable accuracy, while the SVM model shows high variability and lower predictive performance.

The results of this study are consistent with recent trends in the use of artificial intelligence models to assess concrete properties and project performance. For example [56],

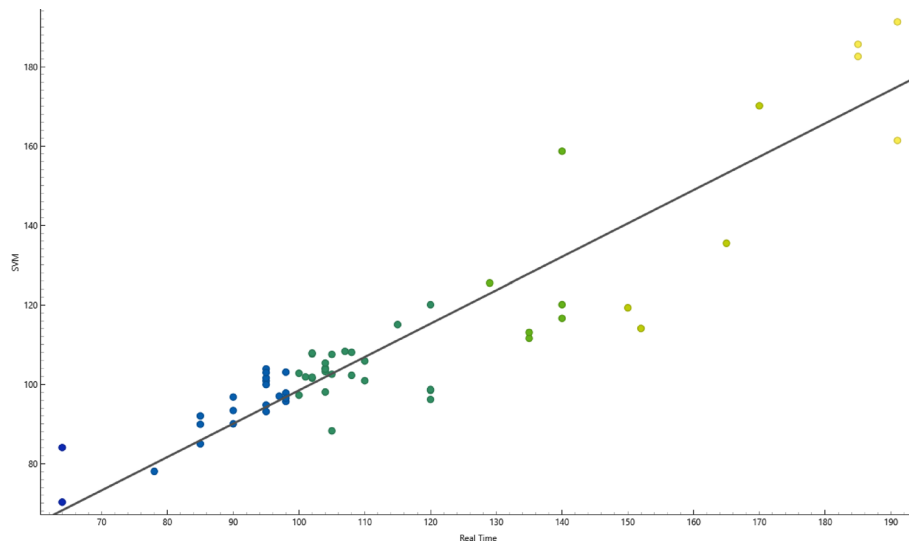


Fig. 4 Scatter plot between actual and predicted values using the SVM model

demonstrated the effectiveness of the Hybrid Random Forest model in predicting the resistance of concrete with high fly ash content, achieving similar performance in terms of prediction accuracy. In another study, machine learning techniques were used to analyse the effect of alternative materials such as manufactured sand and fly ash on concrete resistance, demonstrating the ability of models to handle multiple variables.

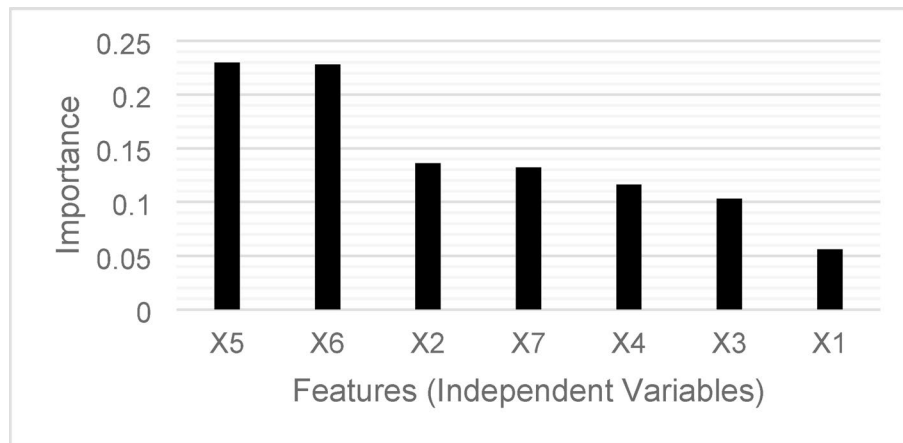
In addition, [57] employed XGBoost and LightGBM algorithms to predict chloride permeability in high-performance concrete and achieved strong results, with XGBoost excelling in explaining the importance of variables.

In comparison, this study shows similar performance results, with the added advantage of using real-world data from housing projects in Iraq, which enhances the applicability of the proposed models. The interpretation of variables through XGBoost is an important factor in supporting planning decisions in the early stages.

5 Feature importance

Understanding the impact of each input variable is a fundamental step in enhancing model interpretation and identifying the most influential factors in construction productivity. By analysing the feature importance extracted from the XGBoost model, variables can be ranked according to their relative impact on the output, contributing to model improvement and supporting practical engineering decisions.

From Fig. 5, the importance of each input can be noticed. The quantity of brickwork (X5) and the quantity of concrete work (X6) have the highest importance values, 0.23 and 0.228, respectively, indicating that the quantity of brickwork and concrete work considerably influences the construction productivity or required time. This is logical since brickwork and concrete are the main elements of housing structural works in Iraq, and the quantity of brickwork directly impacts the labour and time required to finish the project [58]. The third important input was that the built area (X2) directly impacted the required time and positively correlated with the required quantities, especially brickwork quantities. It is noted that the variable number of teams (X7) has a significant impact on the duration of implementation, which is logical given that team size affects the speed of completion. It is worth noting that this variable can be effectively managed through

**Fig. 5** Features (input) Importance**Table 4** Validation results

Validation set	Project	Real	XGBoost	ANN	SVM
1	P11	108	106	125	108
2	P11	101	102	105	102
3	P11	105	104	109	103
4	P11	152	152	160	114
5	P12	90	91	95	97
6	P12	85	85	86	85
7	P12	191	190	187	161

workforce planning and task allocation, thereby opening up the possibility of utilizing the model in human resource management applications and recommendation systems. It should be noted that the concrete teams do not perform concrete work; rather, there are dedicated and independent teams for concrete pouring. This input enables the model to consider the availability and effectiveness of labour in carrying out housing structural works [59]. The built ratio (X4) (built area/total area) shows moderate importance on the outcome. The number of rooms (X3) and the total area (X1) importance values are 0.103 and 0.056, respectively, which means these two inputs (variables) have the lowest impact on the prediction. Each input in the model plays a role in forecasting construction productivity. The significance values represent the relative impact of each component, with brickwork and concrete work quantities and the number of brickwork team members being the most important.

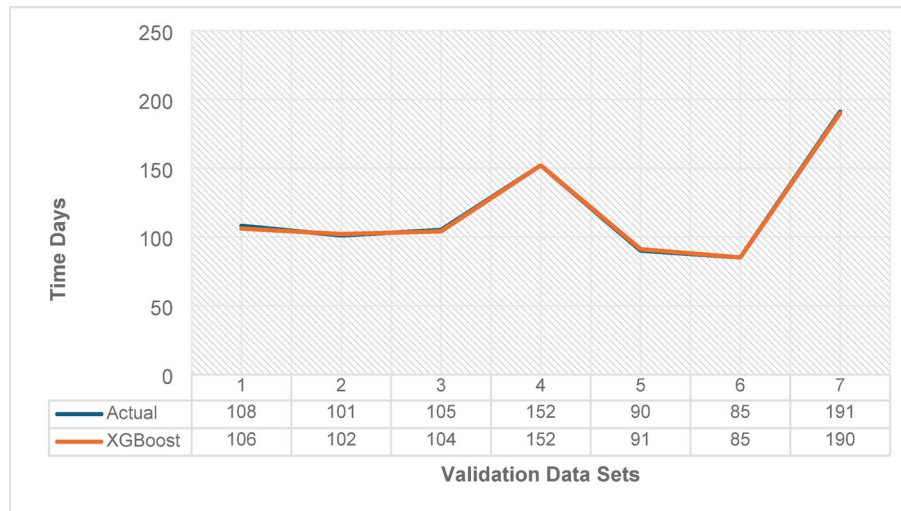
6 Model validation

The models were validated using seven data sets, comparing the actual and predicted values of structural housing works productivity. Table 4 below shows the seven data sets used to validate the three models, which are not the same as those used to develop the model. Notably, the predicted Y and actual Y values are so close that statistical metrics have been calculated to ensure the model's accuracy.

Statistical metrics values shown in Table 5 suggest that the average deviation between the predicted and actual values is slight, indicating a high level of accuracy for the XGBoost model. The model shows good predictive performance with strong correlation

Table 5 Validated model metrics

Parameters	Values
MPE%	0.175
AA%	99.23
Correlation coefficient R	0.99
R ²	0.99

**Fig. 6** The actual and predicted time

and reasonably low error metrics according to MPE%, AA%, and correlation coefficient values.

Figure 6 compares actual values and predicted values using the XGBoost model for seven validation sets. The verification points (1, 2, 3, 4, 5, 6, 7) show a high degree of consistency between the actual time and the expected time, reflecting the model's ability to accurately predict the execution time. This convergence in results reinforces the reliability and stability of the model and supports the statistical indicators obtained previously.

7 Limitations and future work

This study has made a practical contribution by developing predictive models for the productivity of horizontal housing projects in Iraq based on real data from implemented projects, which gives the results high credibility and clear practical value. However, there are several limitations that should be noted, which open the door to future studies:

7.1 Sample size

The database included only 67 horizontal housing projects. Although this number meets the statistical threshold (5–10 observations per independent variable), it remains limited given the complexity of machine learning models. This is because obtaining larger data sets from actual projects is challenging due to the difficulty of documentation and record availability in the construction sector. The small sample size may increase the likelihood of overfitting and reduce the generalisability of the results. In future studies, it is recommended to try to expand the database by collaborating with government and private institutions to collect additional project data, while applying external validation

to independent data to enhance the strength of the models and expand their scope of application.

7.2 Scope of study

The research focused only on horizontal housing projects (detached units), providing an accurate understanding of this type of project. However, the research did not include vertical housing projects (apartments and residential towers), which have different productivity patterns due to the nature of their implementation and labour intensity.

Future studies could include vertical housing projects and compare the different patterns, allowing for a more comprehensive framework for understanding the productivity of housing projects in their various forms.

7.3 Input variables

The models were limited to seven key variables (built area, land area, number of rooms, building ratio, amount of concrete, amount of bricks, and number of brick teams). Despite their importance, other real-world factors affecting productivity, such as worker experience, site management efficiency, material quality, and the level of technology used, were not included. It is therefore suggested that future studies include human, managerial, technical, and economic variables, thereby increasing the accuracy of the models and their ability to explain productivity variation more broadly.

7.4 Practical application

Although the developed models (ANN and XGBoost) showed high accuracy, they have not yet been translated into a practical tool that can be used directly by engineers and planners.

In future studies, it is recommended to develop a graphical user interface (GUI) or a simple digital tool that allows project data to be entered and generates immediate predictions, with field testing in new projects to measure its effectiveness in supporting early planning decisions and resource management.

8 Conclusion

This study demonstrates a clear scientific contribution to the field of predicting the productivity of housing projects in Iraq using modern artificial intelligence techniques. Three predictive models based on machine learning techniques were developed: artificial neural network (ANN), support vector machine (SVM), and XGBoost algorithm, relying on real data extracted from 67 actually implemented housing projects.

The results showed that the XGBoost model performed best in this study, achieving the highest coefficient of determination ($R^2 = 0.998$) and the lowest mean absolute percentage error (MAPE = 0.80%), indicating its high ability to handle complex data and non-linear relationships. It was followed by the ANN model, which demonstrated good performance ($R^2 = 0.92$ and MAPE = 4.1%), reflecting its ability to predict with acceptable accuracy when its structure is adjusted appropriately. The SVM model performed less well than the other two models ($R^2 = 0.813$ and MAPE = 6.6%), which is consistent with previous studies that have addressed the limitations of this model in processing multi-variate construction data.

The variable importance analysis also showed that the quantity of bricks and concrete and the number of work teams had the greatest impact on determining productivity, reflecting the prominent role of practical implementation aspects in determining the time required to complete the work, which is consistent with the reality of horizontal housing projects in Iraq.

These results highlight the potential of machine learning techniques, particularly XGBoost and ANN, as promising tools to support decision-makers in the initial stages of housing project planning. They enable a better understanding of performance rates and a more accurate estimation of time and resources, thereby enhancing project management efficiency. However, the limited sample size (67 projects) is a fundamental limitation, requiring future studies to expand the database and conduct external validation to increase the reliability and generalisability of the results.=

Based on this, these models can be viewed as scalable tools that contribute to improving resource management and reducing time and financial waste in the housing sector. As databases continue to be developed and these models are integrated into practical digital tools, it will be possible to make greater use of them to support efforts to develop the construction sector and contribute to meeting the growing demand for housing in Iraq.

Abbreviations

ANN	Artificial neural network
XGBoost	Extreme gradient boosting
SVR	Support vector regression
CP	Construction productivity
GDP	Gross domestic product
MSE	Mean squared error
RMSE	Root mean squared error
MAE	Mean absolute error
MAPE	Mean absolute percentage error
R ²	Coefficient of determination
R	Correlation coefficient
ML	Machine learning
GUI	Graphical user interface
MIC	Modular integrated construction
CNN	Convolutional neural network
SVM	Support vector machine
FIS	Fuzzy inference system
BNN	Bayesian neural network
HVFA SCC	High-volume fly ash self-compacting concrete
GGBFS	Ground granulated blast furnace slag
ANN-GOA	Artificial neural network–Grasshopper optimization algorithm

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Author contributions

Ali Hassoon led the research, including developing the study concept, collecting and analysing data, developing models, and writing the first draft of the manuscript. Dr. Farid Izani Mohamed Ghazali and Dr. Tareq A. Khaleel contributed by supervising the research process, providing methodological guidance, verifying the models' results, and critically reviewing the manuscript. All authors reviewed and approved the final version of the research and acknowledge full responsibility for all aspects of the work.

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Data availability

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Generative AI tools

ChatGPT, version 4.0, was used to improve the language.

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